

Date	03 July 2026
Team ID	
Team Size	3
Team Leader	Chakali Basavaraju
Team Member 1	Girish C
Team Member 2	Shaik Sameer
Team Member 3	
Team Member 4	
Project Name	Smart Lender: Machine Learning-Based Loan Approval Prediction System
Maximum Marks	

Technology Stack

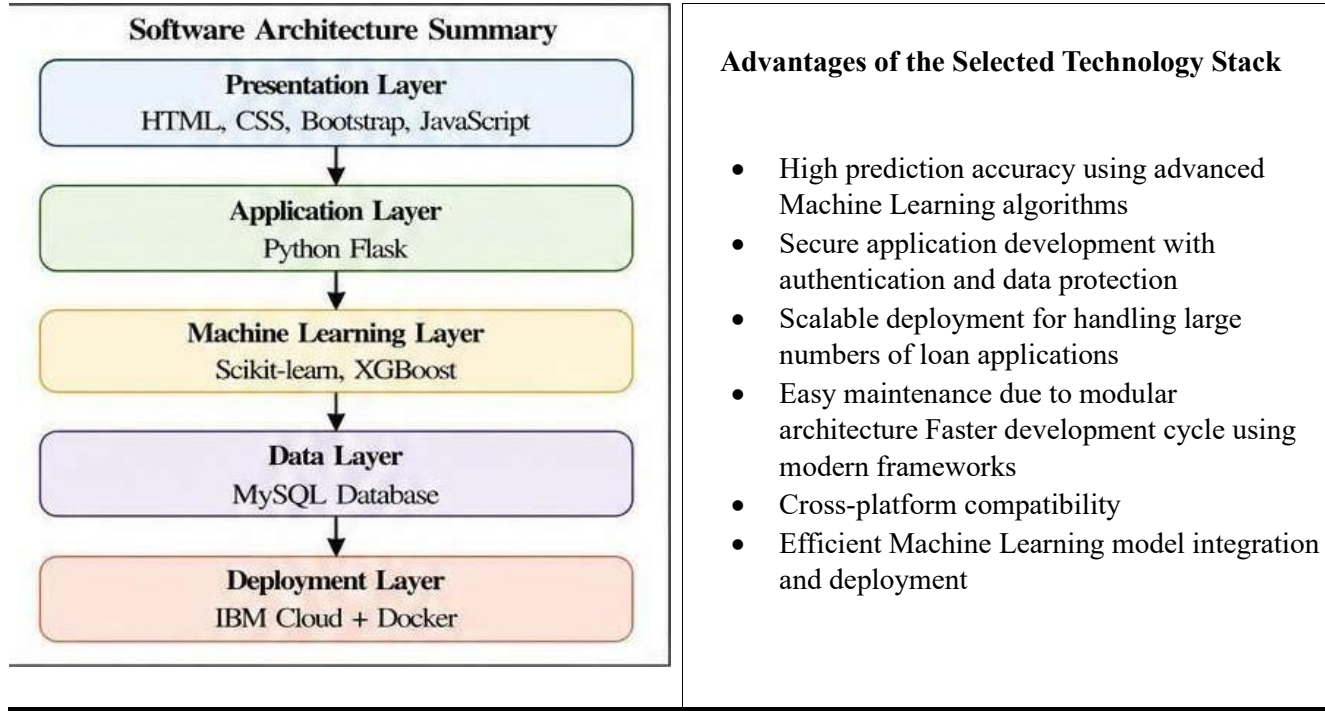
Introduction

	Technol(W	
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript	Designs responsive user interfaces for loan applicants and administrators.
Backend	Python, Flask	Implements application logic, API routing, authentication, and loan prediction services.
Database	MySQL	Stores user accounts, applicant details, prediction history, and system records.
Machine Learning	Scikit-learn, XGBoost, Pandas, NumPy	Supports data preprocessing, feature engineering, model training, evaluation, and prediction.
Data Visualization	Matplotlib, Seaborn	Generates analytical charts and dataset visualizations.

Development Environment	Visual Studio Code, Jupyter Notebook	Used for development, experimentation, debugging, and model training.
-------------------------	---	---

Technology Summary

The selected technology stack provides a strong foundation for building an intelligent loan approval prediction system. It supports efficient data processing, accurate Machine Learning predictions, responsive user interfaces, and secure data handling. These technologies enable seamless integration between frontend, backend, database, and Machine Learning components.



Conclusion

The selected technology stack combines modern web development technologies, powerful Machine Learning frameworks, reliable database systems, and scalable deployment solutions. Together, these technologies enable the Smart Lender Loan Approval Prediction System to deliver secure, scalable, maintainable, and accurate loan approval predictions suitable for real-world financial applications. The stack provides a robust foundation for building an intelligent, efficient, and user-friendly system.

The selected technologies provide a robust foundation for developing an intelligent, secure, scalable, and maintainable Machine Learning—Based Loan Approval Prediction System.

Technology Stack (Continued)

Layer	Technol(40'	
Cloud Deployment	IBM Cloud, Docker	Deploys the trained Machine Learning model and web application with scalable cloud infrastructure and containerization.
Version Control	Git, GitHub	Tracks source code changes, supports collaboration, manages versions, and maintains project documentation.
Model Storage	Joblib	Stores and loads trained Machine Learning models efficiently for prediction.
API Testing		Tests REST APIs and validates application endpoints.
Testing frame work	PyTest	Performs functional and integration testing to improve software quality.
Operating System	Windows 11	Provides the primary software development environment.
Virtual Environment	Python venv	Manages isolated project dependencies and package versions.
Documentation	Markdown, Microsoft Word	Used for project documentation, technical reports, and GitHub README preparation.